

# NumPy Analyzer

## Professional Project Documentation for Institute Submission

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### Project Overview

NumPy Analyzer is a Python-based project developed using Object-Oriented Programming and the NumPy library. It performs array operations, mathematical calculations, statistical analysis, filtering, sorting, reshaping, and aggregation functions.

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### Features

- Create 1D and 2D Arrays
  - Insert and Delete Elements
  - Search and Sort Arrays
  - Filter Arrays
  - Mathematical Operations
  - Combine and Reshape Arrays
  - Aggregation Functions
  - Statistical Functions
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### Technologies Used

- Python
  - NumPy
  - Object-Oriented Programming (OOP)
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### Project Structure

```
NumPy_Analyzer/  
■■■ numpy_analyzer.py  
■■■ README.md  
■■■ requirements.txt
```

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### Run Command

```
python3 numpy_analyzer.py
```

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## Conclusion

This project demonstrates practical implementation of Python programming and NumPy functionalities through a menu-driven application suitable for educational and analytical purposes.